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THE ROLE OF DEEP LEARNING IN CYBER DECEPTION TECHNIQUES FOR NETWORK DEFENSE

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ABSTRACT

In the ever-evolving landscape of cybersecurity, the efficacy of traditional defense mechanisms against sophisticated threats is increasingly questioned. Cyber deception, a strategy rooted in deliberately misleading adversaries, has gained prominence as a proactive defense approach. Concurrently, deep learning, a subset of artificial intelligence, has emerged as a potent tool for analyzing vast data sets and detecting intricate patterns. This paper explores the fusion of deep learning and cyber deception techniques for network defense. Drawing from historical contexts and contemporary cybersecurity challenges, we investigate how deep learning enhances deception strategies, thereby augmenting network defense capabilities. Through an analysis of case studies, we illustrate the practical application of deep learning in cyber deception scenarios. Additionally, we address challenges such as data scarcity, interpretability issues, and adversarial attacks, while proposing future research avenues. This study provides insights into the synergy between deep learning and cyber deception, offering valuable perspectives for advancing network defense strategies in the digital age.

Keywords: Cybersecurity, Network Defense, Deep Learning, Cyber Deception, Artificial Intelligence, Threat Detection, Anomaly Detection, Adversarial Attacks, Data Privacy, Network Security.

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1. INTRODUCTION TO CYBER DECEPTION AND NETWORK DEFENSE

Cyber deception and network defense have become critical components in the ever-evolving landscape of cybersecurity. As organizations increasingly rely on interconnected systems and digital infrastructure, the threat landscape continues to expand, with adversaries employing sophisticated techniques to infiltrate networks, compromise sensitive data, and disrupt operations. In response, cybersecurity professionals are continuously seeking innovative strategies to bolster their defense mechanisms and mitigate the risks posed by cyber threats.

Cyber deception refers to the deliberate manipulation of information to mislead adversaries, divert their attention, and ultimately thwart their malicious activities within a network environment. Unlike traditional security measures that focus primarily on detection and prevention, cyber deception involves the proactive deployment of deceptive techniques and decoys to confuse, delay, or deter attackers. These techniques may include the creation of fictitious assets, fake credentials, and deceptive communication channels, all designed to lure attackers into a simulated environment where their actions can be monitored, analyzed, and neutralized.

Network defense, on the other hand, encompasses a broad range of strategies, technologies, and practices aimed at safeguarding the integrity, confidentiality, and availability of networked systems and data. From firewalls and intrusion detection systems to encryption protocols and access controls, network defense measures are implemented to detect, prevent, and respond to unauthorized access, malware infections, and other cyber threats. However, the traditional approaches to network defense often fall short in detecting and mitigating advanced threats, such as targeted attacks and insider threats, which require more proactive and adaptive defense mechanisms.

In recent years, the integration of cyber deception techniques into network defense strategies has emerged as a promising approach to augment traditional security measures and enhance the overall resilience of organizations against cyber threats. By leveraging the principles of deception, organizations can create dynamic and adaptive defense environments that not only detect and respond to threats in real-time but also actively deceive and disrupt adversaries' activities. This proactive approach not only reduces the dwell time of attackers within the network but also provides valuable insights into their tactics, techniques, and procedures (TTPs), enabling organizations to better understand their adversaries and strengthen their defense posture accordingly.

In this research paper, we aim to explore the role of deep learning in advancing cyber deception techniques for network defense. Deep learning, a subset of artificial intelligence (AI) that mimics the human brain's neural networks, has shown remarkable capabilities in processing vast amounts of data, identifying patterns, and making complex decisions with minimal human intervention. By harnessing the power of deep learning algorithms, cybersecurity researchers and practitioners can develop more sophisticated and effective deception techniques that adapt to the evolving threat landscape and outsmart even the most determined adversaries.

Throughout this paper, we will examine the evolution of cyber deception in network defense, discuss the principles and applications of deep learning in cybersecurity, explore case studies and examples of deep learning-driven deception techniques, analyze the challenges and limitations of deep learning in network defense, and propose future directions and emerging trends in leveraging deep learning for cyber deception. By gaining a deeper understanding of the intersection between deep learning and cyber deception, organizations can better prepare themselves to defend against the increasingly sophisticated and persistent cyber threats they face in today's digital world.

2. THE ROLE OF DEEP LEARNING IN CYBERSECURITY

Cybersecurity is a rapidly evolving field that faces an ever-growing number of sophisticated threats. As organizations strive to protect their digital assets and sensitive information from cyberattacks, they are increasingly turning to advanced technologies such as deep learning to bolster their defense capabilities. Deep learning, a subset of artificial intelligence (AI) that mimics the human brain's neural networks, has emerged as a powerful tool for addressing the complex and dynamic nature of cyber threats. In this section, we explore the role of deep learning in cybersecurity and examine its applications across various domains.

Enhancing Threat Detection and Prevention: One of the primary applications of deep learning in cybersecurity is in threat detection and prevention. Deep learning algorithms can analyze vast amounts of data, including network traffic, logs, and system activity, to identify patterns indicative of malicious behavior. By training deep neural networks on labeled datasets of known threats, cybersecurity professionals can develop highly accurate models capable of detecting and blocking suspicious activities in real-time. These models can help organizations proactively defend against a wide range of cyber threats, including malware, phishing attacks, and insider threats.

Improving Anomaly Detection: Traditional cybersecurity approaches often rely on predefined rules and signatures to detect anomalies in network traffic or user behavior. However, these methods may struggle to identify novel or previously unseen threats. Deep learning techniques, such as recurrent neural networks (RNNs) and convolutional neural networks (CNNs), excel at learning complex patterns and relationships from data, making them well-suited for anomaly detection tasks. By training deep learning models on historical data, organizations can build robust anomaly detection systems capable of identifying subtle deviations from normal behavior that may indicate a security breach.

Strengthening Authentication and Access Control: Deep learning can also play a crucial role in strengthening authentication and access control mechanisms. By analyzing user behavior patterns, biometric data, and contextual information, deep learning algorithms can enhance traditional authentication methods such as passwords and tokens, making them more secure and resilient to attacks. Furthermore, deep learning models can be used to detect and prevent unauthorized access attempts in real-time, helping organizations protect their sensitive data and resources from unauthorized users.

Automating Threat Response: In addition to detection and prevention, deep learning can also be used to automate threat response and remediation efforts. By integrating deep learning models with security orchestration, automation, and response (SOAR) platforms, organizations can automate incident response workflows and streamline the process of identifying, containing, and mitigating security incidents. Deep learning-powered systems can analyze security alerts, prioritize incidents based on their severity and potential impact, and initiate predefined response actions without human intervention, enabling organizations to respond to threats more rapidly and effectively.

Adapting to Evolving Threats: One of the key advantages of deep learning in cybersecurity is its ability to adapt to evolving threats and changing attack techniques. Traditional cybersecurity solutions often struggle to keep pace with the rapidly evolving threat landscape, requiring frequent updates and manual intervention to remain effective. Deep learning models, however, can continuously learn from new data and adapt their behavior over time, allowing them to detect and respond to emerging threats more effectively. By leveraging deep learning techniques, organizations can build more resilient and adaptive cybersecurity defenses capable of defending against both known and unknown threats.

The deep learning holds great promise for transforming cybersecurity by enabling organizations to detect, prevent, and respond to cyber threats more effectively. By leveraging the power of deep learning algorithms, organizations can enhance threat detection and prevention capabilities, improve anomaly detection, strengthen authentication and access control mechanisms, automate threat response, and adapt to evolving threats in real-time. As the cybersecurity landscape continues to evolve, deep learning is likely to play an increasingly important role in helping organizations stay ahead of emerging threats and protect their critical assets from cyberattacks.

3. EVOLUTION AND APPLICATION OF DECEPTIVE TECHNIQUES

Deceptive techniques have long been employed as a strategy in warfare and intelligence operations, but their application in cybersecurity, particularly within the realm of network defense, has undergone significant evolution in recent years. This evolution is driven by the increasing sophistication of cyber threats and the recognition that traditional defense mechanisms alone are insufficient to thwart determined adversaries. In this section, we delve into the historical evolution and modern-day application of deceptive techniques in cybersecurity.

3.1. Historical Evolution:

Deceptive techniques have been used throughout history to mislead adversaries and gain strategic advantages in conflict situations. In traditional warfare, tactics such as camouflage, decoys, and misinformation have been employed to confuse, divert, or deceive enemy forces. Similarly, in espionage and intelligence operations, deception has been used to manipulate perceptions, conceal true intentions, and gather valuable information from adversaries.

In the context of cybersecurity, the use of deceptive techniques can be traced back to the early days of computing, where techniques such as honeypots and honeytokens were employed to lure and trap attackers. These early forms of deception aimed to create fake assets or bait within a network environment to attract and identify malicious actors. While these techniques were effective to some extent, they lacked sophistication and were often easily bypassed by skilled attackers.

3.2. Modern-day Application:

In recent years, advancements in technology, particularly in the fields of artificial intelligence and machine learning, have enabled the development of more sophisticated and effective deceptive techniques in cybersecurity. These techniques leverage the principles of deception to create dynamic, adaptive, and realistic environments that mimic legitimate systems and data, thereby fooling attackers into revealing their intentions and tactics.

One of the key applications of deceptive techniques in cybersecurity is in the realm of threat detection and attribution. By deploying decoys, traps, and breadcrumbs strategically throughout a network environment, organizations can create virtual minefields that deter attackers and expose their presence. Deception can also be used to gather valuable intelligence about adversaries, such as their tactics, techniques, and procedures (TTPs), which can then be used to enhance defensive measures and mitigate future threats.

Another application of deceptive techniques is in the field of incident response and threat hunting. By using deception to lure attackers into predefined engagement zones, organizations can gather real-time intelligence about ongoing attacks and initiate response actions to mitigate their impact. Deception can also be used to proactively hunt for threats within a network environment, identifying hidden or dormant adversaries before they have a chance to strike.

Furthermore, deceptive techniques can be used to protect critical assets and data by creating virtual fortresses around sensitive resources. By surrounding valuable assets with layers of deception, organizations can deter attackers and make it more difficult for them to locate and compromise their targets. Deception can also be used to disrupt attackers' reconnaissance efforts and delay their progress within a network environment, giving defenders more time to detect and respond to threats.

The evolution and application of deceptive techniques in cybersecurity represent a paradigm shift in the way organizations approach network defense. By embracing deception as a proactive and strategic defense strategy, organizations can turn the tables on attackers and gain the upper hand in the ongoing battle against cyber threats. However, it is important to recognize that deception is not a panacea and should be used in conjunction with other defensive measures to create a comprehensive security posture. As cyber threats continue to evolve, the role of deceptive techniques in cybersecurity is likely to become increasingly important in deterring, detecting, and mitigating attacks in real-time.

4. DEEP LEARNING IN CYBER DECEPTION: CASE STUDIES

Case studies serve as valuable illustrations of how deep learning techniques are effectively applied in the realm of cyber deception, showcasing their practical implementation and real-world impact. In this section, we present several compelling case studies that demonstrate the use of deep learning in cyber deception strategies, highlighting their effectiveness in enhancing network defense and thwarting malicious activities.

- i. **Adversarial Machine Learning for Deceptive Traffic Generation:** In this case study, researchers leveraged deep learning techniques to generate deceptive network traffic designed to confuse and mislead attackers. By training deep neural networks on large datasets of legitimate network traffic, the researchers developed models capable of generating synthetic traffic that mimicked the behavior of real users and devices. These synthetic traffic patterns were then strategically injected into the network environment, creating a camouflage effect that made it more difficult for attackers to differentiate between legitimate and malicious activities. The results demonstrated that the use of deep learning-generated deceptive traffic significantly increased the complexity and uncertainty for attackers, reducing their effectiveness and increasing the likelihood of detection.
- ii. **Deep Learning-Driven Honeypots for Threat Intelligence Gathering:** Honeypots have long been used as a deception technique to lure attackers into a controlled environment where their actions can be monitored and analyzed. In this case study, researchers enhanced traditional honeypot deployments by integrating deep learning techniques for more intelligent threat intelligence gathering. By analyzing the vast amounts of data collected from honeypot interactions, deep learning models were trained to identify patterns indicative of malicious behavior and classify attackers based on their TTPs. This enabled organizations to gain valuable insights into emerging threats and attacker techniques, allowing them to better understand their adversaries and strengthen their defense strategies accordingly.
- iii. **Deep Learning-Powered Deception in Endpoint Security:** Endpoint security is a critical aspect of network defense, as endpoints serve as primary targets for attackers seeking to gain access to sensitive data and systems. In this case study, researchers developed a deep learning-powered endpoint deception framework designed to detect and deter malicious activities at the endpoint level. By deploying lightweight agents on endpoints, the framework continuously monitored system behavior and used deep learning algorithms to detect anomalies indicative of suspicious activity. When suspicious behavior was

detected, the framework dynamically generated deceptive responses, such as fake system alerts or decoy files, to confuse and distract attackers. The results demonstrated that the use of deep learning-driven deception at the endpoint level significantly improved detection rates and reduced the dwell time of attackers within the network environment.

- iv. **Dynamic Deception Orchestrated with Deep Learning:** In this case study, researchers explored the use of deep learning techniques to orchestrate dynamic deception campaigns tailored to specific threat scenarios. By analyzing historical attack data and leveraging advanced deep learning algorithms, researchers developed predictive models capable of anticipating attacker behavior and dynamically adjusting deception strategies in real-time. This enabled organizations to proactively adapt their deception tactics to evolving threats, maximizing their effectiveness and minimizing the risk of detection. The results demonstrated that the use of deep learning-driven dynamic deception significantly enhanced organizations' ability to defend against sophisticated and persistent cyber threats.

These case studies highlight the diverse range of applications for deep learning in cyber deception, from generating deceptive traffic and gathering threat intelligence to enhancing endpoint security and orchestrating dynamic deception campaigns. By leveraging the power of deep learning algorithms, organizations can develop more sophisticated and effective deception strategies that outsmart even the most determined adversaries, ultimately strengthening their overall defense posture and mitigating the risk of cyber attacks.

5. CHALLENGES AND LIMITATIONS OF DEEP LEARNING

While deep learning has shown remarkable promise in various fields, including cybersecurity, it is not without its challenges and limitations. One significant challenge is the insatiable appetite for data that deep learning algorithms exhibit. Deep learning models typically require large volumes of labeled data to train effectively, and obtaining such datasets in the cybersecurity domain can be challenging due to the scarcity of labeled cyber threat data. Furthermore, labeled data may not always be representative of the diverse and evolving nature of cyber threats, leading to potential biases in the trained models.

Another challenge is the interpretability and explainability of deep learning models. Deep neural networks are often described as "black boxes" due to their complex architectures and opaque decision-making processes, making it difficult for cybersecurity professionals to understand how and why a particular decision was made. This lack of transparency can hinder trust and adoption, as organizations may be reluctant to rely on automated systems that they cannot fully comprehend. Additionally, the inability to interpret the inner workings of deep learning models makes it challenging to diagnose and correct errors or biases that may arise during training or deployment.

Scalability is another limitation of deep learning in cybersecurity. While deep learning models have demonstrated impressive performance on specific tasks, such as image recognition or natural language processing, scaling these models to handle the complexity and volume of data encountered in cybersecurity applications can be prohibitively resource-intensive. Training deep neural networks on large-scale cybersecurity datasets requires significant computational power and memory resources, which may not be readily available to all organizations. Furthermore, deploying deep learning models in real-world cybersecurity environments can pose challenges in terms of latency, throughput, and resource consumption, particularly in resource-constrained environments such as edge devices or IoT devices.

Robustness and adversarial attacks are also significant concerns in deep learning cybersecurity applications. Deep neural networks have been shown to be vulnerable to adversarial attacks, where malicious actors can deliberately manipulate input data to deceive the model and induce

incorrect predictions. Adversarial attacks can have serious consequences in cybersecurity applications, potentially leading to false positives or false negatives that compromise the security of the system. Developing robust and resilient deep learning models that are resistant to adversarial attacks remains an ongoing research challenge in the field of cybersecurity.

The ethical and privacy considerations are important limitations to consider when deploying deep learning models in cybersecurity applications. Deep learning algorithms have the potential to capture and analyze vast amounts of sensitive information, raising concerns about data privacy, consent, and surveillance. Organizations must carefully consider the ethical implications of using deep learning techniques in cybersecurity, ensuring that they adhere to relevant regulations and standards governing data protection and privacy. Additionally, bias and discrimination in deep learning models can perpetuate existing inequalities and exacerbate societal issues, underscoring the need for transparency, fairness, and accountability in the development and deployment of deep learning algorithms.

While deep learning holds great promise for revolutionizing cybersecurity, it is essential to recognize and address the challenges and limitations associated with its use. By understanding these challenges and working to overcome them, cybersecurity professionals can harness the full potential of deep learning to develop more effective and resilient defense mechanisms against evolving cyber threats.

6. FUTURE DIRECTIONS IN DEEP LEARNING FOR NETWORK DEFENSE

As the cyber threat landscape continues to evolve rapidly, fueled by advancements in technology and the increasing sophistication of malicious actors, the role of deep learning in network defense is poised to undergo significant expansion and innovation. In this section, we explore potential future directions in deep learning for network defense, outlining emerging research areas, technological advancements, and strategic considerations that are likely to shape the future of cybersecurity.

- i. **Adversarial Machine Learning and Robustness:** A key focus of future research in deep learning for network defense will be on addressing the robustness and resilience of deep learning models against adversarial attacks. Adversarial machine learning techniques, which aim to systematically exploit vulnerabilities in deep learning models, have emerged as a significant threat to the security and reliability of AI systems. Future research efforts will focus on developing robust and resilient deep learning architectures that are capable of defending against adversarial attacks while maintaining high levels of accuracy and performance.
- ii. **Explainable AI and Model Interpretability:** Another area of focus will be on enhancing the interpretability and explainability of deep learning models in network defense applications. As deep learning models become increasingly complex and opaque, there is a growing need for methods and techniques that enable cybersecurity professionals to understand and interpret the decisions made by these models. Future research will explore techniques for explaining the underlying rationale behind deep learning predictions, identifying potential biases and vulnerabilities, and enhancing the transparency and trustworthiness of AI systems.
- iii. **Transfer Learning and Domain Adaptation:** Transfer learning and domain adaptation techniques will play a critical role in the future of deep learning for network defense. Transfer learning enables the transfer of knowledge learned from one task or domain to another, allowing deep learning models to leverage pre-trained representations and adapt them to new environments or scenarios. Future research will explore methods for effectively transferring knowledge between related cybersecurity tasks, domains, or

datasets, enabling more efficient and scalable training of deep learning models for network defense.

- iv. **Federated Learning and Privacy-Preserving AI:** Federated learning and privacy-preserving AI techniques will also be increasingly important in the future of deep learning for network defense. Federated learning enables the collaborative training of deep learning models across multiple decentralized devices or data sources without sharing raw data, thereby preserving privacy and confidentiality. Future research will focus on developing federated learning algorithms and protocols that are tailored to the unique requirements and constraints of network defense applications, enabling organizations to leverage distributed data sources while maintaining data privacy and security.
- v. **Self-Learning and Autonomous Cyber Defense:** A long-term vision for the future of deep learning in network defense is the development of self-learning and autonomous cyber defense systems. These systems would be capable of continuously monitoring, analyzing, and adapting to evolving cyber threats in real-time, without human intervention. Future research will explore techniques for enabling deep learning models to learn autonomously from their interactions with the network environment, dynamically adjusting their defense strategies to counter emerging threats and vulnerabilities.
- vi. **Multi-Modal Learning and Fusion:** Multi-modal learning and fusion techniques will enable deep learning models to leverage diverse sources of information, such as network traffic data, log files, sensor data, and contextual information, to enhance network defense capabilities. Future research will explore methods for integrating information from multiple modalities and sources, enabling deep learning models to capture richer and more comprehensive representations of the network environment and detect subtle anomalies or patterns indicative of malicious activity.

The future of deep learning for network defense holds great promise for addressing the evolving challenges and complexities of cybersecurity. By embracing emerging research areas, technological advancements, and strategic considerations, cybersecurity professionals can harness the full potential of deep learning to develop more robust, resilient, and adaptive defense mechanisms against a wide range of cyber threats. However, achieving this vision will require interdisciplinary collaboration, innovative research, and a continued commitment to advancing the state-of-the-art in deep learning for network defense.

7. CONCLUSION

In this paper, we delved into the integration of deep learning techniques within cyber deception strategies for network defense. We initially acknowledged the persistent challenge of defending against sophisticated cyber threats, prompting the exploration of innovative approaches such as cyber deception. Through an exploration of deep learning's role in this context, we demonstrated its potential to significantly enhance network defense strategies. Deep learning algorithms, with their capacity to analyze extensive data sets, detect intricate patterns, and make real-time decisions, are well-suited to address the dynamic nature of cyber threats.

Our examination, illustrated through case studies, showcased how deep learning can effectively generate deceptive traffic, gather threat intelligence, bolster endpoint security, and orchestrate dynamic deception campaigns. Despite these advancements, we also underscored various challenges and limitations associated with deep learning in cybersecurity, including data scarcity, interpretability issues, scalability concerns, susceptibility to adversarial attacks, and ethical considerations.

Our findings emphasize the critical role of deep learning in revolutionizing network defense through cyber deception. By leveraging deep learning algorithms, organizations can enhance traditional security measures, thwart adversaries, and adapt to evolving threats more effectively. The key takeaways include the necessity of addressing challenges and limitations, continued research, and collaboration to advance deep learning's application in cybersecurity. As organizations navigate the complexities of the digital landscape, embracing deep learning in network defense becomes paramount for safeguarding critical assets and information against cyber threats.

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